08 - Consumer Trust Analytics

Survey instrument, scoring, statistics

Knowledge-pack rule: every statement below is derived from files in this project or is explicitly labelled as planned, missing or not verified. Nothing here is an invented citation, dataset, result or feature.

Source files used for this document

survey/questionnaire.md; survey/survey_data.csv; src/trust_scoring/scoring.py; src/model_evaluation/statistics.py; src/preprocessing/anonymize.py

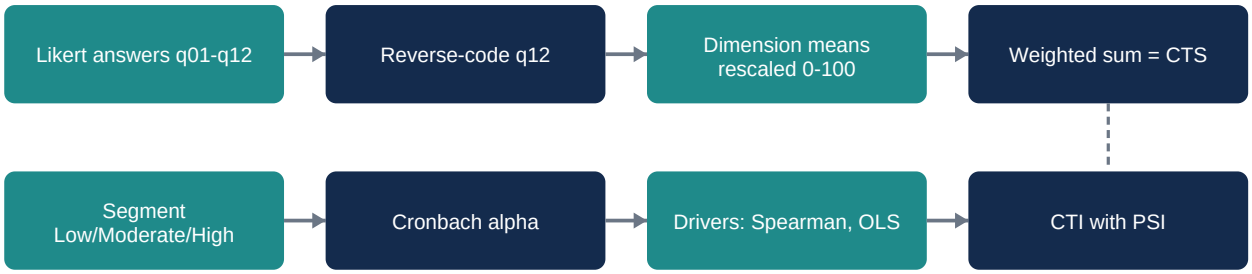
The bundled survey_data.csv is SYNTHETIC (latent-factor generator). Its statistics demonstrate the method and must not be reported as findings about real consumers.

Instrument

Item	Statement	Dimension	Reverse
q01	I feel my money is safe when I use my main FinTech app.	perceived_security	no
q02	My FinTech provider uses strong security measures (e.g. MFA, encryption).	perceived_security	no
q03	I am confident my FinTech provider can detect and stop fraud.	perceived_security	no
q04	My FinTech provider handles my personal data responsibly.	privacy_assurance	no
q05	I understand how my financial data is used and shared.	privacy_assurance	no
q06	My provider clearly communicates security incidents and policy changes.	transparency	no
q07	Security and privacy settings are easy to find and understand.	transparency	no
q08	The app works reliably when I make payments.	reliability	no
q09	My provider resolves fraud or dispute cases quickly and fairly.	reliability	no
q10	I would recommend my FinTech provider to friends or family.	behavioural_intent	no
q11	I intend to keep using FinTech services for my main finances.	behavioural_intent	no
q12	I worry that my FinTech account could be hacked. (reverse-coded)	perceived_security	yes

Behaviour and demographic fields: age_band, education, usage_frequency, uses_mfa, experienced_fraud, digital_literacy. No free text.

How the score is computed



Implemented in src/trust_scoring/scoring.py.

Quantity	Definition
CTS	sum of weight x dimension score / sum of weights
PSI	100 x mean(recall, 1 - min(10 x FPR, 1), attack block rate, validation pass rate)

Quantity	Definition
CTI	0.6 x mean CTS + 0.4 x PSI

CTI and PSI are the project's own proposals. Their weights are design choices; no psychometric validation (factor analysis, test-retest) has been performed.

Actual results from the last run (synthetic survey)

Quantity	Value
Respondents	355
Mean CTS	56.57
PSI	79.75
CTI	65.84
OLS R-squared (MFA, past fraud, literacy)	0.360
Smallest k-anonymity group	5

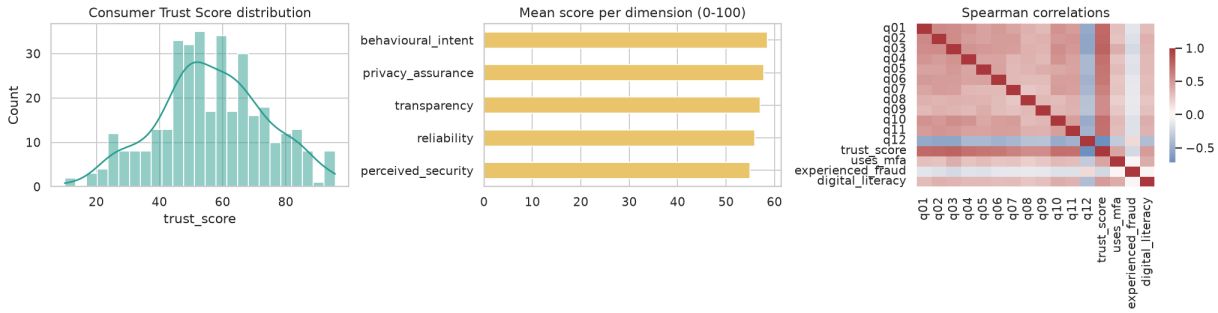
Dimension	Mean (0-100)
perceived_security	55.0
privacy_assurance	57.8
transparency	57.1
reliability	56.0
behavioural_intent	58.5

Segment	Respondents
Low	122
Moderate	178
High	55

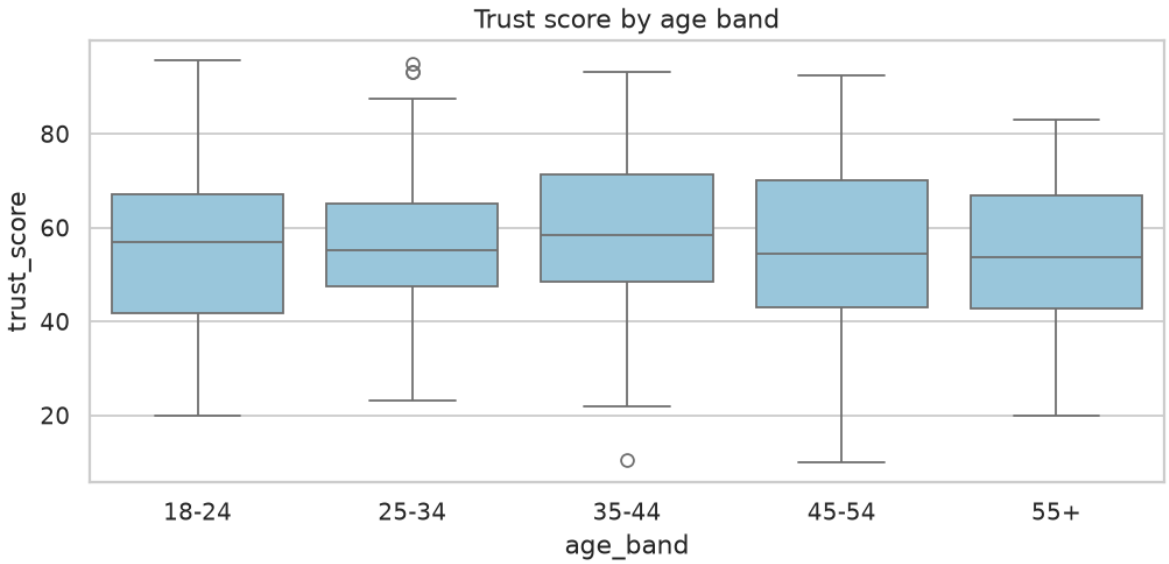
Scale	Cronbach alpha
overall	0.906
perceived_security	0.838
privacy_assurance	0.601
transparency	0.517
reliability	0.490
behavioural_intent	0.647

Driver	Spearman rho with CTS	p-value
digital_literacy	0.478	1.07e-21
uses_mfa	0.431	1.86e-17
experienced_fraud	-0.224	2.02e-05

The directions (MFA positive, past fraud negative) are built into the synthetic generator, so agreement with them confirms the pipeline recovers the planted structure; it is not discovery about people.



Trust score distribution, dimension means and Spearman correlations (generated by the pipeline).



Trust score by age band (synthetic).

Privacy handling (implemented)

- Salted-hash pseudonyms (SURVEY_SALT); age banded; no direct identifiers.
- k-anonymity (k=5) check and suppression: src/preprocessing/anonymize.py.

Gaps

Item	Status
Real respondents / consent records	MISSING
Ethics approval record	MISSING
Pilot study for wording and alpha	PLANNED
Factor analysis / index validation	PLANNED
Sensitivity analysis notebook (05)	IMPLEMENTED